

IT353 BLOCKCHAIN TECHNOLOGY



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Module - IV

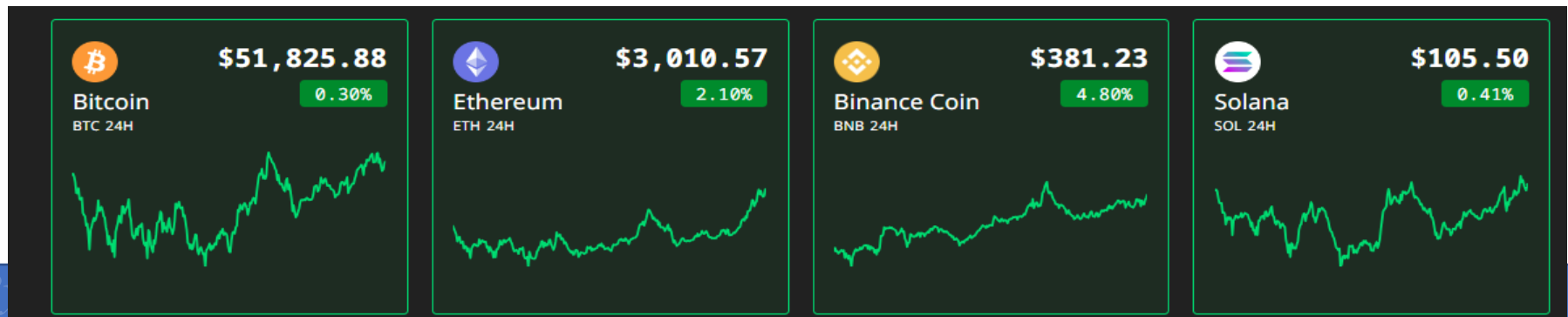
- **Bitcoin and Security:**
- Bitcoin creation, exchanges. Wallets, security.
- Protecting blockchain from attackers.
- Forks soft and hard,
- Blockchain security,
- Key Management in Bitcoin,
- Case studies.

Introduction

- **Digital Currency** is the digital format of the physical currency that we carry around in wallet or withdraw from an ATM and is backed by an authority (RBI).
- **Cryptocurrency** is not backed by a central authority and may be used to trade for digital assets but may lack legal tender.

Digital Currency	Cryptocurrency
Centralized	Decentralized
E-cash	Secure cryptographically
Hard to achieve transparency	Transparent flow
Stable	Unstable
Legal	Legality issues

- There are many cryptocurrency existing



Introduction

- **Bitcoin** is the first cryptocurrency introduced in 2009 by Satoshi Nakamoto.
- It is a consensus network that enables a new payment system and a completely digital money.

- **Characteristics of Bitcoin**

- Decentralized
- Transparent
- Fast
- Immutable
- Volatile



Bitcoin Calculator

Crypto to FIAT

Crypto to Crypto

1

BTC

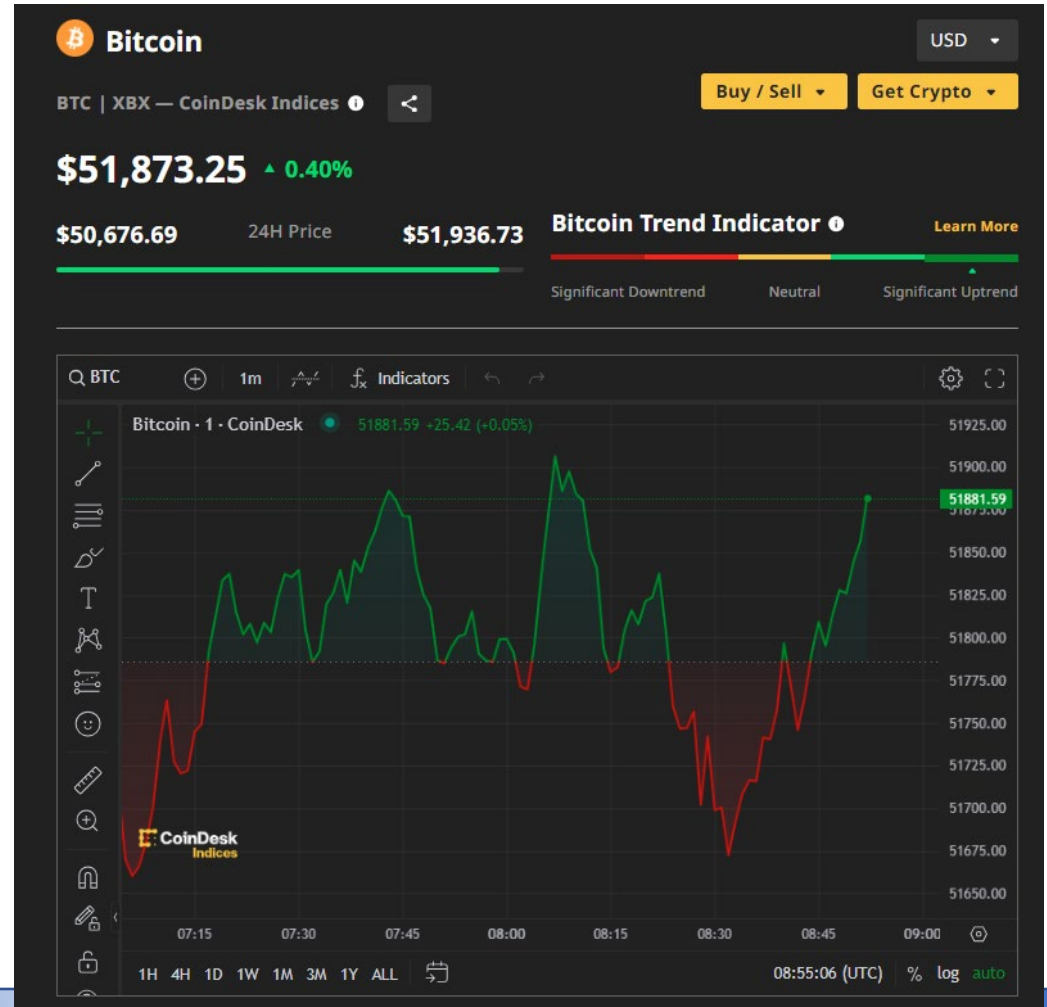
Bitcoin

51,873.25

USD

United States Dollar

Buy / Sell



Bitcoin - Working

Bitcoin Wallet

- Bitcoin wallet is a device or program that is used to send and receive bitcoins.
- A Bitcoin wallet holds at least one “account,” or sub-wallet.
- Each Bitcoin “account” within a Bitcoin wallet has information associated with it. The two key pieces of information are the public bitcoin address and the private key.
- The public address is comparable to the debit card’s account number. The private key, meanwhile, is kind of like a debit card’s password in that it grants access to the bitcoin associated with that public bitcoin address.
- One of the key functions of a Bitcoin wallet is to manage the private key which is 256 bit number.
- Bitcoin wallets provide a way to write down this private key in a much more human readable format, referred to as a recovery phrase, secret passphrase, or seed phrase. A recovery phrase is a list of words, usually between 12 and 24, that allow you to reconstruct your Bitcoin wallet and gain access to your funds even if your Bitcoin wallet is destroyed.

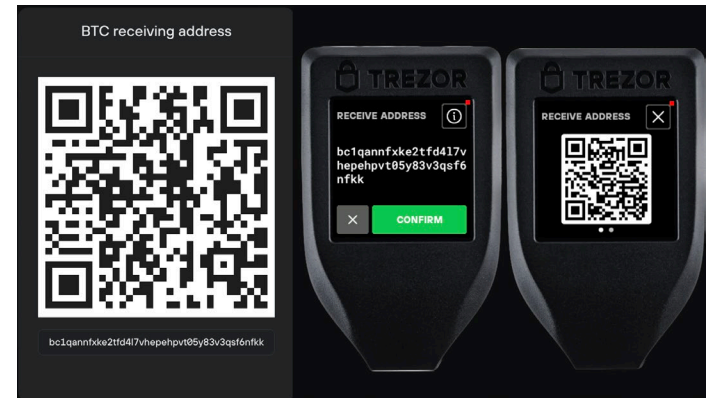
Bitcoin - Working

Bitcoin Wallet

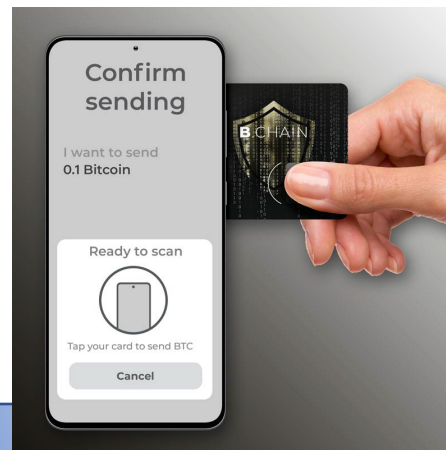
- There are two main types of wallets: hardware wallets and software wallets.
 - Hardware Bitcoin wallets are the most secure type of Bitcoin wallet because their private keys are stored on a physical device. When a person wishes to make a transaction, they plug in the hardware wallet (usually via USB). Ledger and Trevor are well known hardware wallets.



<https://coinfomania.com/ledger-wallet-announces-important-security-risk/>



<https://trezor.io/learn/a/receive-crypto-in-trezor-suite>

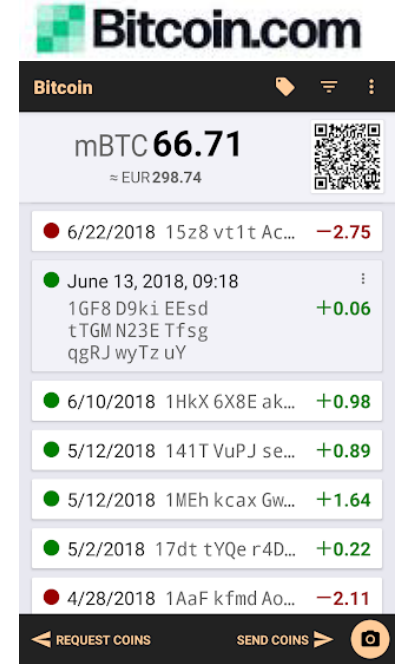
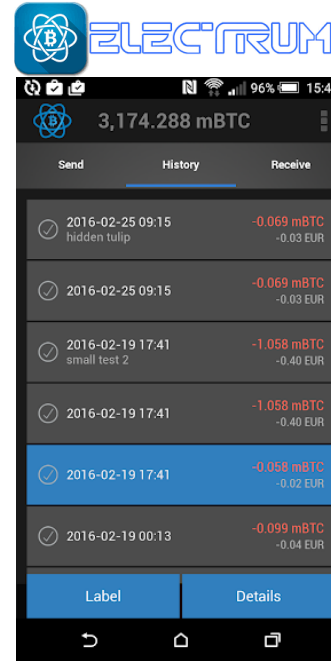
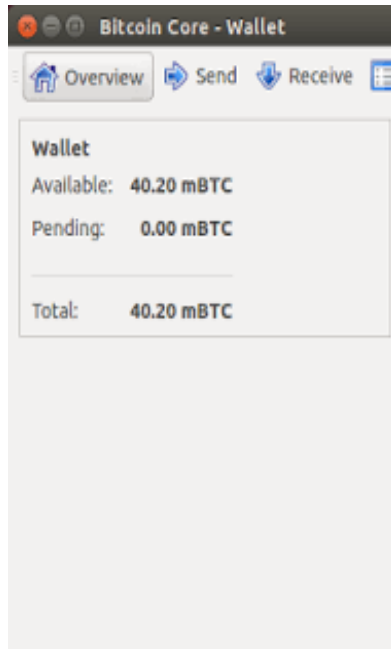
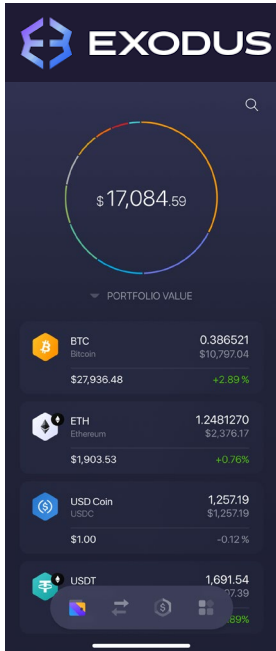


<https://www.idemia.com/card-based-crypto-hardware-wallet>

Bitcoin - Working

Bitcoin Wallet

- Software Bitcoin wallets exist on computing devices, like a desktop computer, a smartphone, or in a web browser.



Bitcoin - Working

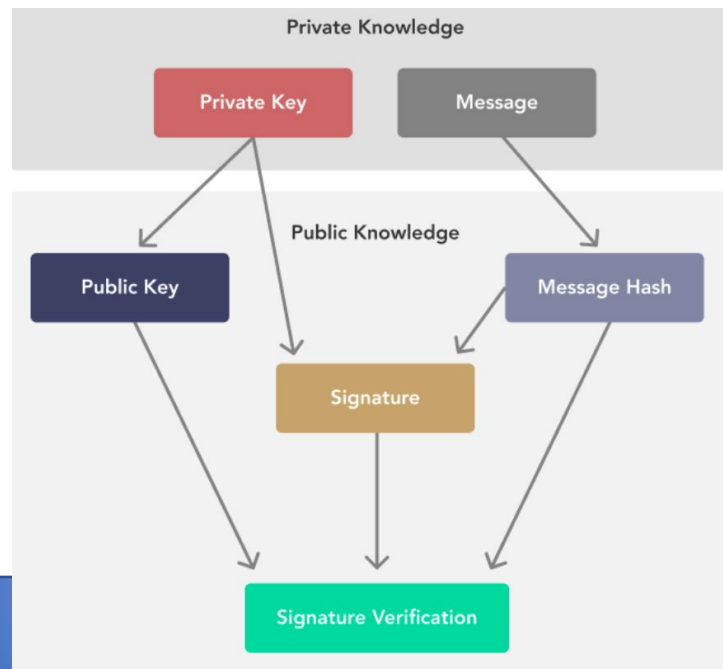
Bitcoin Key Generation

- Cryptocurrency and other digital assets are not in the wallet but on the blockchain (distributed ledger).
- These assets are assigned to the wallet by means of a key through which it can be accessed.
- A 'key pair' consists of a public key (an address) and the corresponding unique private key (a password).
- Public keys, also called bitcoin addresses, are randomly generated sequences of letters and numbers that function similarly to an email address or a social-media site username.
- The private key is another sequence of letters and numbers which are like passwords to email or other accounts, are to be kept secret.
- Formally, a private key for Bitcoin is a series of 32 bytes. Bitcoin uses Public-Key Cryptography to encrypt and decrypt data for authenticity and privacy.
- Private keys are randomly generated 256-bit hexadecimal numbers that is hashed using SHA-256 algorithm (Secure Hash Algorithm with 256 bit length). This creates a private key in a hexadecimal format.
- Public keys are generated mathematically using Elliptic curve which is a mathematical process where the private key multiplies by a variable in an elliptic curve to produce an output. The generated output is then hashed using SHA-256 to produce a 256 bit output, followed by RIPEMD160 (RACE Integrity Primitives Evaluation Message) to produce a 160-bit length output. This process is known as double hashing.

Bitcoin - Working

Creating a Transaction

- When we send a bitcoin, we create a transaction from our digital wallet.
- A transaction is a transfer of value between Bitcoin wallets that gets included in the block chain.
- This transaction includes the sender's address (public key), the recipient's address (public key), the amount of Bitcoin to be sent, and a transaction fee.
- Private keys are used to create signatures. A signature is a piece of data that can only be created or calculated using the private key.
- However, the validity of a signature can be verified using only the public key.
- This feature is used to ensure that bitcoin can only be spent by its rightful owner, who controls the corresponding private keys.
- The signature also prevents the transaction from being altered by anybody once it has been issued.



Bitcoin - Working

Creating a Transaction

- Many bitcoin wallets allow users to set transaction fees manually. The fees are directly sent to the miners.
- Bitcoin users can control how quickly their transactions are processed by setting the fee rate. The higher the fee rate, the faster the transaction will be processed.
- Miners are incentivized to prioritize the transaction with the highest fees. During times of high network congestion, where a large number of users want to transact, the transactions with the highest fees are more likely to be included in the next block.
- All Bitcoin transactions are published to the mempool, where they are considered 'pending'. When a miner adds a transaction to a block, it is then considered confirmed.

Mining

- All transactions are broadcast to the network and usually begin to be confirmed within 10-20 minutes, through a process called mining.
- Mining is a distributed consensus system that is used to confirm pending transactions by including them in the block chain.
- To be confirmed, transactions must be packed in a block that fits very strict cryptographic rules that will be verified by the network.
- These rules prevent previous blocks from being modified because doing so would invalidate all the subsequent blocks.

Bitcoin - Working

Mining

- A user who generates a transaction will broadcast his proposed transaction to the Bitcoin network.
- A special group of participants in the network known as 'miners' verify that the user's keys are able to access the inputs.
- Miners also gather together a list of other transactions that were broadcast to the network around the same time (from the Mempool) and form them into a block.
- Any miner who has completed the Proof of Work is permitted to propose a new block that will be added to the chain. This miner also receives the transaction fees for the confirmed transactions.
- That new block is then broadcast to the network. If other network participants (nodes) agree it's a valid block (ie. the transactions it contains follows all the rules of the protocol and it properly references the previous block), they will pass it along.
- Only 1 megabyte of transaction data can fit into a single Bitcoin block as limited by the creators.

Bitcoin - Security

- Authentication

- Am I paying the right person? Not some other impersonator?
→ Public Key Crypto: Digital Signatures

- Integrity

- Is the coin double-spent?
- Can an attacker reverse or change transactions?
→ Digital Signatures and Cryptographic Hash / Timestamp

- Availability

- Can I make a transaction anytime I want?
→ Distributed Ledger

- Confidentiality

- Are my transactions private? Anonymous?
→ Pseudonymity using addresses (key)

Protecting blockchain from attackers

- Phishing attack

- Phishing is a scamming attempt to attain a user's credentials.
- Fraudsters send wallet key owners emails designed to look as though they're coming from a legitimate source.
- The emails ask users for their credentials by using fake hyperlinks. Having access to a user's credentials and other sensitive information can result in losses for the user and the blockchain network.

- Routing attacks

- Blockchains rely on real-time, large data transfers.
- Hackers can intercept data as it's transferring to internet service providers.
- In a routing attack, blockchain participants typically can't see the threat, so everything looks normal. However, behind the scenes, fraudsters have extracted confidential data or currencies.

- Sybil attacks

- In a Sybil attack, hackers create and use many false network identities to flood the network and crash the system.
- An attacker creates multiple fake nodes on the network. Using those nodes the attacker can obtain majority consensus and hamper the transactions on the chain.

- 51% attacks

- If a miner, or a group of miners, might rally enough resources, they might attain more than 50% of a blockchain network's mining power. Having more than 50% of the power means having control over the ledger and the ability to manipulate it.

Protecting blockchain from attackers

- Eclipse Attack:

- Hackers try to make duplicates of the node. This is an eclipse attack. The user eclipses(hides) the original node and broadcasts the fake node that was created by the hacker.

Security solutions

- Use of Cold Wallet
- Using other consensus mechanism instead of Proof of Work can help prevent the 51% attacks
- Reputation based consensus works well against Sybil attacks.
- It is important to keep a check on the mining pools and limit them to divert miners to other pools.
- The use of secure routing protocols (one with certificates) can help prevent routing attacks on the blockchain.
- Avoid public Wi-Fi networks.
- Smart contracts need to be thoroughly examined by experts before implementation.
- Blockchain Penetration Testing by ethical hackers.
- Using private permissioned blockchain wherever possible.

Forks in Blockchain

- Forks in blockchain means copying the code and modifying it to create a new software or product.
- In open-source projects Forks are very common and used widely. So, cryptocurrencies like Ethereum and Bitcoin are decentralized and open software so that anyone can contribute.
- Forks can make user interface more interactive and look good.
- Forks can be Codebase or live.
- Live blockchain forking may be Temporary or Permanent.
- Temporary forks happens during mining of a block by more than one party at nearly the same time and is eventually resolved later.
- Permanent forks can be soft or hard forks.
- In soft fork new rules can be added such that they do not clash with the old rules. When there is a change in the software that runs on the nodes to function as a network participant, the change is such that the new blocks mined on the basis of new rules are also considered valid by the old version of the software. This feature is also called as backwards-compatibility.
- When there is a change in the software that runs on the full nodes to function as a network participant, the change is such that the new blocks mined on the basis of new rules are not considered valid by the old version of the software, it is known as hard fork.
- When hard forks occur, new currency come into existence like in the case of Ethereum (original : Ethereum, new : Ethereum Classic) and Bitcoin (original : Bitcoin, new : Bitcoin cash).
- Equivalent quantity of currency is distributed to the full nodes who choose to upgrade their software so that no material loss occurs

Forks in Blockchain

Why do we need forks

- Add new functionality
- Fix security issues